SEMINAR SUMMARY:
JANUARY 12, 2026 - DAVID CHAN
WHAT IS RESEARCH?

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What is Research? The process of learning enough information to sufficiently answer a question.

In Tuesday January 12th's seminar, David Chan opened the talk with a summary of the course generally. We began by noting that every summary document assignment (due the Friday following the seminar) should contain the following elements:

- the name of the presenter,
- the presentation title, and
- a thorough but consise summary.

The class expectations follow as:

- be on time,
- pay attention,
- ask questions.

The goals for the course in general follow as:

- to discover the research interests of a collection of faculty members,
- to experience a fairly broad range of research topics,
- to experience a broad range of presentation styles, and finally
- to gain a better understanding of what research is and how to do it.

When doing guided research with a professor there are a few expectations that go along with the hours obtained. The following give a fairly comprehensive set of expectations for research operation:

(1) work consistently,
(2) meeting regularly,
(3) explain your results to others (for example in seminars),
(4) training (e.g. lab policies, software),
(5) reading,
(6) getting frustrated a lot.

Some general advice follows as:

- pick something that you are passionate about,

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- try to find a professor that you are comfortable with,
- pick something that is in an area of strength
- **think on your own**
- enjoy yourself!

After this description of the courses logistics and goals Dr. Chan got into his individual areas of research, which we'll address in the following sections.

1. Project 1. Rippled Dynamics

The idea at hand here is to study these sequence generators mainly for the purpose of modeling drug interactions with epilepsy patients. Though the lens through which we are looking at these drug interactions is differnce equations in the discrete dynamnical system field. The curiousity with regards to these sequence generating functions is that they are both chaotic and periodic with their behaviour.

The model that we were given follows as

$$x_{n+1} = \max\left\{\frac{A_n + D_n H_\epsilon(x_n)}{x_n}, \frac{B_n}{x_n - 1}\right\}$$

where A_n and D_n are periodic generating sequences and $H_\epsilon(x)$ is the heavyside function.